

ROCK PAPER SCISSORS

A real-time computer vision system that plays Rock Paper Scissors against you.

RASPBERRY PI 5

MEDIPIPE AI

WEBRTC

WEB-BASED

DawsonCodes

SENIOR PROJECT / 2026

// 01 THE WHY

Why I Chose This Project

FUTURE PLANS

I'm going to Michigan State University in the fall to study computer science with a focus on AI and machine learning. I wanted a senior project that would actually push me toward that field instead of being something easy.

INSPIRATION

I saw a YouTuber build something similar and thought "I could do that." I've built smaller projects before but never something this complete — hardware, ML, networking, and a custom interface in one system.

THE CHALLENGE

Rock Paper Scissors is simple as a game, but to make a computer play it well you need real-time hand tracking, gesture classification, low-latency video, and a clean interface. Lots of separate problems.

// 02 MENTOR

Project Mentor

MENTOR

My dad

(Dad)

Honest framing: he didn't write code with me, but mentored the parts of the project that weren't code.

HARDWARE & FUNDING

Helped pick out the Pi 5, AI Hat+, camera, and cooling. Covered the hardware costs so I could focus on building.

MORAL SUPPORT & FEEDBACK

When I hit walls — and I hit a lot — he'd talk me through what was working and what wasn't. Kept me on track.

// 03 BEGINNING

How I Began

RASPBERRY PI 5

Latest single-board computer. Quad-core ARM, 8GB RAM.

AI HAT+ (HAILO-8)

26 TOPS neural processor. Built for vision AI.

NEXIGO N60 USB CAMERA

1080p webcam. The eyes of the system.

ORIGINAL PLAN

Run hand detection on the Hailo NPU. Use the AI hat for actual AI.

// 04 EARLY WINS

What Went Well Early On



LINUX SETUP

Pi 5 OS install, drivers, and SSH access — all working in under an hour.



CAMERA WORKS

USB camera detected on first try. Linux saw the video device immediately.



COMPUTE POWER

Pi 5 plus AI Hat+ is serious hardware. Way more than I thought I'd need.

// 05 COMPLICATIONS

First Walls I Hit

HAILO NPU TOOLCHAIN

Required converting the model to a special HEF format. Documentation was scattered. Burned hours fighting it.

PYTHON MEDIAPIPE FAILED

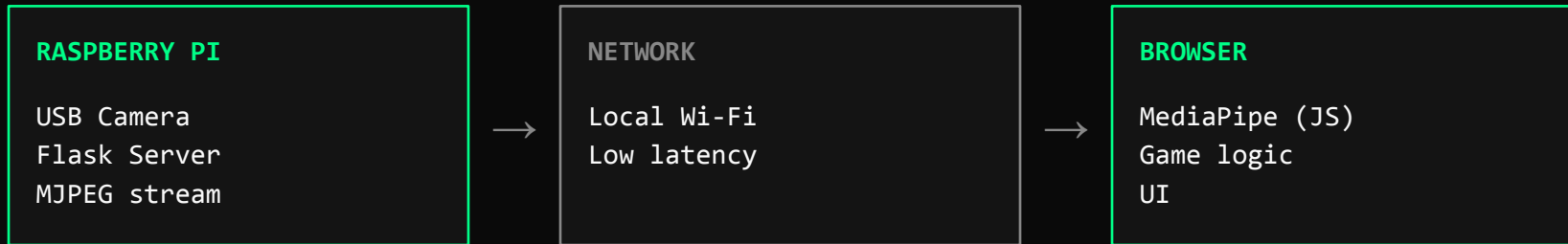
Pi was running Python 3.13. MediaPipe didn't have a build for it. "No matching distribution found."

DECISION POINT

Two paths blocked. First time I had to make a real engineering decision instead of following a tutorial — what do I give up on?

// 06 PIVOT

Pivot: Move the AI to the Browser



Pi just sends video. The browser does the AI. Any device with a modern browser can play.

// 07 COMPLICATIONS

Three Problems at Once

MJPEG WAS LAGGY

Each frame is a full JPEG. No compression between frames. Visible delay when my hand moved fast.

BAD GESTURE CLASSIFIER

My homemade fingertip-distance algorithm worked for clean poses, fell apart at any angle.

UI LOOKED ROUGH

Cluttered, small camera feed, weird colors. Felt like a school project.

// 08 WINS

Middle of Project — Wins

Multi-Device

PHONE / LAPTOP / TABLET

13 FPS

BROWSER INFERENCE RATE

End-to-End

FULL GAME LOOP WORKING

REAL PROGRESS

Even with the lag and rough UI, the core idea worked. Network streaming was real. Hand detection was real. The game cycle ran end to end. The bones of a working system were there.

// 09 MENTOR CHECK-IN

Reality Check from Dad

"It's not broken. You're iterating. That's what engineers actually do."

– dad

REFRAMED THE PROBLEM

Iteration is the process. Each version was further along than the one I'd given up on.

PERMISSION TO SCRAP

Stop being afraid to delete code. Sunk cost is real. That conversation saved the project.

// 10 ACCOMPLISHMENT

Switching to WebRTC

BEFORE: MJPEG

- ✗ Full JPEG every frame
- ✗ No compression between frames
- ✗ Noticeable lag

AFTER: WEBRTC

- ✓ Real video compression
- ✓ Only sends what changed
- ✓ Used by Zoom & Google Meet

Cost: complete server rewrite. Flask → aiohttp + aiortc. Worth every minute.

// 11 COMPLICATIONS

New Problems With WebRTC

CPU-HEAVY ENCODING

Software encoding caps the framerate around 25 fps. Hardware encoding would be a much bigger rewrite.

ICE NEGOTIATION

Devices have to figure out how to talk to each other. Even on local Wi-Fi it can stall. Added a timeout.

BROWSER CACHING

Updates wouldn't show up — browser kept loading old code. Learned about hard refresh (Ctrl+Shift+R) the hard way.

// 12 SUCCESS

Streaming Felt Instant

Lag dropped from “annoying” to “unnoticeable.”

FOUNDATION FOR EVERYTHING ELSE

Streaming was no longer the bottleneck. Any remaining problems had to be in detection or UI. That clarity let me focus on one problem at a time.

// 13 REFLECTION

Honest Check-In

TIRED BUT MOTIVATED

Some days I'd work for hours and feel like I made nothing. Closed the laptop more frustrated than I opened it.

STARTING TO ENJOY IT

Once streaming was solid, the project flipped from grinding to addictive. Every fix showed up immediately on screen.

THIS IS WHAT I WANT

It clicked that the work I want to do for a living is exactly this. Solving messy problems where the answer isn't in a textbook.

// 14 MENTOR ADVICE

Dad: “Don’t Try to Do Everything”

“Finish what’s 80% there. A polished demo beats a half-finished feature dump.”

ROBOT ARM → FUTURE WORK

My original plan included a physical robot hand. Cut from this version, kept on a future work list.

POLISH OVER FEATURES

Changed my whole approach for the home stretch. Quality of what's there beats quantity of what's added.

// 15 HOME STRETCH

The Home Stretch

01

BETTER DETECTION

Scrapped my custom algorithm. Switched to MediaPipe's GestureRecognizer — a real ML model trained on hand gestures.

02

LOCK-HOLD SYSTEM

You have to hold a pose steady for a moment before it commits. Eliminated almost all false positives.

03

POLISHED UI

Redesigned everything. Dark theme. Floating overlays. Win/loss flash. Looks like a real product.

// FINAL BUILD

// 16 FINAL FEATURES

Finishing the Project

IMPOSSIBLE MODE

Computer always wins. Reads your move in real time, picks the counter when you press play. Demo magic.

SKELETON OVERLAY

Debug view drawing the 21 hand landmarks. You can see exactly what the AI sees on your hand.

POLISHED UI

Dark theme. Floating overlays. Win/loss flash. Hotkey panel. Feels like a real app, not a school project.

// 17 FINAL BUGS

Bugs Hunted in the Home Stretch

AI HALLUCINATIONS

Sometimes detected a “hand” that wasn't a hand — a fold in my shirt, a wall pattern. Added a size sanity check.

DEMO CHEAT EXPLOIT

Found a way to cheat my own Impossible Mode. Lock a move, drop your hand, press play fast. Patched with a freshness check.

INVERTED WIN LOGIC

Game was sometimes telling me I won when I should have lost. Found because Impossible Mode wasn't impossible enough.

// 18 FINAL WINS

What I Shipped

\$80

TOTAL HARDWARE COST

100%

WEB-BASED / NO APP INSTALL

Multi-User

ANY DEVICE / SAME WI-FI

IT WORKS, IT LOOKS REAL, IT RUNS ON A PI

Detection is accurate. Streaming is smooth. UI is polished. Impossible Mode does exactly what it's supposed to. Anyone walking up to it would think it's a polished product.

// 19 THE NUMBERS

Overall — Project Wrap

1-2 mo

TOTAL TIME / WORKED IN CHUNKS

~800

LINES OF CODE / PY + JS + HTML/CSS

3

COMPLETE ARCHITECTURE REWRITES

Three rewrites sounds like failure. It's the part I'm proudest of.

I didn't get it right the first time. Most real engineering doesn't.

// 20 RESEARCH

Research That Got Me There

MEDIAPIPE DOCS

Studied how the GestureRecognizer task works. Learned the 21 hand landmarks the model returns and how to run it via WebAssembly.

WEBRTC + AIORTC

Read about ICE, SDP offers, peer connections. Enough to ship a working version, not enough to claim I understand all of WebRTC.

PI 5 HARDWARE

Researched the Pi 5 chip and what 26 TOPS actually means. Even unused, that knowledge shaped my decisions.

// 21 REFLECTION

How I Feel Now

PROUD

Three rewrites, late nights, a lot of doubt. And it came together. I have something I can demo, not just an assignment.

MORE CONFIDENT

Going into MSU this fall. Before this I knew I liked CS. Now I know I can finish a real piece of engineering.

NOT DONE YET

Robot arm. Hailo NPU. Multi-camera. This is the school version. The real version is what's next.

LIVE DEMO

Time to play.

01

FAIR MODE

A few rounds with random CPU moves.
Show the detection working and the
score updating.

02

SKELETON OVERLAY

Press D. The 21 landmark points appear
on my hand in real time.

03

IMPOSSIBLE MODE

Press I. I try to win. Watch the score
climb against me. I cannot win.

Thank you. / Questions welcome. / Anyone can play afterward.